

Algorithms HW3

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1, 24, 12

#22

13	2^n	8	$n - n^2 + 5n$	11	$(\frac{3}{2})^n$
2	$\lg n$	9	n^3	6	$n \lg n$
8	$\frac{n}{2}$	4	$\sqrt{n}$	1	b
7	$(\lg n)^2$	5	n^3		
14	$n!$	10	$n + \lg n$		
3	$\lg \lg n$	12	$2^n - 1$		

#23

a	$g \in O(f)$
b	$f \in O(g)$
c	$f \in O(g)$
d	$f \in O(g)$
e	$f \in O(g)$
f	$f \in O(g)$
g	$g \in O(f)$
h	$f \in O(g)$

#7a

$$3 \sum_{i=1}^n i + 7 \sum_{i=1}^n$$

use formula

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$$7e. \sum_{i=1}^n 6^i = \frac{6^{n+1} - 6}{5}$$

7b.

$$\sum_{i=1}^n i^2 - 2 \sum_{i=1}^n i$$

$$7f. 4 \sum_{i=1}^n - 4 \sum_{i=0}^n$$

7c

$$\sum_{i=7}^n 4^i = \sum_{i=0}^n 4^i - \sum_{i=0}^6 4^i$$

7d

$$\sum_{i=5}^n (2i^2 + i) \\ 2 \sum_{i=5}^n i^2 + \sum_{i=5}^n i$$